

PL/SQL Packages (Self-Learning)

Q. Discuss the concept of Packages in database programming (such as PL/SQL). Explain the difference between a Package Specification and a Package Body, and list their advantages.

[DEF] Definition to Remember

A Package (in Oracle PL/SQL) is an advanced database programming construct that logically groups related Stored Procedures, Functions, Variables, Cursors, and Exceptions into a single named module. It is analogous to a "Class" or "Namespace" in object-oriented programming, consisting of a Package Specification (public interface) and a Package Body (implementation).

1. Structure of a Package

A package is strictly divided into two separate components:

PACKAGE SPECIFICATION (Declares: procedures, functions, variables, constants, cursors)	<- PUBLIC Interface (What the package can do)
PACKAGE BODY (Contains actual executable code + private variables/procedures)	<- PRIVATE Implementation (How it does it)

A. Package Specification (Public Interface)

B. Package Body (Implementation)

2. Advantages of Packages

Advantage	Explanation
Modularity	Groups related routines (e.g., all HR
Encapsulation	Body hides private logic; only spec is
Performance	Entire package loaded into memory on f
Overloading	Same procedure/function name can have

[MUST-WRITE] Must-Write Points (for 10 marks)

1. A Package groups related procedures, functions, variables, and cursors into one module.
2. It has two parts: Package Specification (public interface) and Package Body (implementation).
3. Specification declares the procedure/function signatures -- the public API.
4. Body contains the actual code + private variables not visible in the specification.
5. Performance: entire package loads into memory on first call -- faster subsequent access.
6. Encapsulation: users see only the specification; body logic is hidden (information hiding).
7. Overloading: same name can have multiple versions with different parameter lists.

[RECALL] Quick Recall

Package -> Groups related DB objects -> Specification (Public API) + Body (Private Implementation) -> Advantages: Modularity + Encapsulation + Memory Performance + Overloading